Jinu Nyachhyon

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Education

Advanced College of Engineering and Management — Kathmandu, Nepal
Institute of Engineering (IOE), Tribhuvan University
B.E. in Computer Engineering, Rank: 2 / 96 — 2023

- **Advisor**: Prof. Dr. Subarna Shakya
- **Research area**: Recommender System and Machine Learning

Publications

1. **Nyachhyon J**, Khatiwada A, Timilsina B. NepPlural: A Human-Anchored Benchmark for Persona Annotation in Nepali Migration Discourse. PlurVA-LLM Workshop, AACL-IJCNLP, 2026. [DOI]
2. Wang Y, Sikka H, Mathur Y*, Zhou T*, **Nyachhyon J***, Guruprasad P. GUI-Perturbed: Domain Randomization Reveals Systematic Brittleness in GUI Grounding Models. AABA4ET NeurIPS '26 Workshop. *(*Equal Contribution)* [DOI]
3. Recchia G, Mangat C, **Nyachhyon J**, Sharma M, Canavan C, Epstein-Gross D, Abdulbari M. Confirmation bias: A challenge for scalable oversight. AAAI-AIA, 2026. [DOI]
4. **Nyachhyon J**, Sharma M, Thapa P, Bal BK. Consolidating and Developing Benchmarking Datasets for the Nepali Natural Language Understanding Tasks. AACL-IJCNLP, 2025. [DOI]
5. Thapa P*, **Nyachhyon J***, Sharma M*, Bal BK. Development of Pre-Trained Transformer-based Models for the Nepali Language. (CHiPSAL) COLING, 2025. *(*Joint First Author)* [DOI]

Talks & Presentations

- *Consolidating and Developing Benchmarking Datasets for the Nepali Natural Language Understanding Tasks*, **AACL-IJCNLP 2025**, Conference Presentation.
- *Development of Pre-Trained Transformer-based Models for the Nepali Language*, **COLING 2025**, Oral Presentation.

Research Experience

Institute for Research and Innovation in Intelligent Systems (IRIIS) — Nepal
Co-founder & Researcher — 2026.02 - Present

- **Agentic Benchmark Framework**: Designing a formal evaluation framework for LLM agents with an explicit episode schema and multi-dimensional scoring, moving beyond single-number task success rates.
- **Multilingual Planning Evaluation**: Constructing a benchmark for multistep instruction planning over real procedural domains, with gold solutions stored as partial-order DAGs to credit valid alternative orderings.

Manifold Research Group — USA (Remote)
Open Science Research Fellow — 2025.10 - 2026.04

- **Software Control Agents**: Built and benchmarked LLM-driven software control agents on Mind2Web, conducting systematic comparisons of models (e.g., UI-TARS, Qwen2.5-VL) in complex, real-world web environments.
- **Trajectory Generation**: Developed trajectory generation and perturbation pipelines for training and stress-testing agents, enabling robustness evaluation under distribution shifts and unseen task variants.
- **Analysis of Agent Behavior**: Conducted systematic analysis of agent behavior, focusing on spatial reasoning, tool use, and action grounding failures.

Information and Language Processing Research Lab (ILPRL) Kavre, Nepal
Research Assistant, Supervisor: Dr. Bal Krishna Bal 2024.07 - 2025.12

- **Benchmarking:** Led the design and release of the NLUE benchmark (12 Nepali NLU tasks), establishing a standardized evaluation suite for low-resource LLMs.
- **Nepali Corpus for Foundation Models:** Assembled and processed 27.5GB Nepali text corpus addressing data scarcity.
- **Pre-Training and Analysis:** Pretrained BERT/RoBERTa/GPT-2 with instruction tuning, achieving SOTA on Nep-gLUE; conducted comparative evaluation and error analysis.

Modulo Research Ltd. Cambridge, UK (Remote)
Research Intern, Supervisor: Dr. Gabriel Recchia 2024.06 - 2024.08

- **Automated Analysis:** Automated analysis of 20-minute screen recordings, refining scene change detection with 80% precision via URL change tracking and LLM-based event and detail extraction of video frames.
- **LLM Alignment Evaluation:** Conducted scalable oversight experiments by comparing LLM outputs with human annotations to assess and improve intent alignment.

Professional Experience

Insyde AI Maryland, USA (Remote)
AI Developer 2025.01 - Present

- **LLM Agent Systems:** Developed production-grade LLM agents for financial workflows with multi-step reasoning, tool use, and automated decision pipelines.
- **MCP & Memory:** Integrated Model Context Protocol (MCP) servers, external tools, and persistent memory for context-aware assistants.
- **Workflow Automation:** Reduced customer processing time by >90%, increasing operational throughput by over 10×.

Virtly IT & Business Solutions Sarl (ICEBRKR) Geneva, Switzerland (Remote)
ML Engineer 2024.03 - 2024.09

- **Finetuning:** Finetuned BART, T5, Pegasus for summarization (ROUGE-L > 50); Phi-3 for task prioritization system.
- **Scheduling Algorithm:** Built an algorithm to resolve online meeting scheduling conflicts by proposing optimal time slots.

LogicTronix Lalitpur, Nepal
ML/CV Engineer 2023.06 - 2024.03

- **3D Object Detection & Tracking:** Integrated and optimized SFA3D into ADAS stack, achieving real-time 3D object detection and tracking on embedded platforms.
- **Model Quantization:** Reduced model size by over 60% through quantization for Xilinx FPGA deployment, improving edge inference efficiency.

Academic Experience

Herald College Kathmandu, University of Wolverhampton Kathmandu, Nepal
Assistant Lecturer 2026.03 - Present

- **Teaching Delivery:** Deliver tutorials and workshops spanning classical deep learning through to modern agentic AI systems, integrating mathematical foundations with hands-on implementation.
- **Practical Guidance:** Design and supervise implementation-based learning (e.g., MNIST and Devanagari datasets, agent-based automation builds), mentoring group and individual work emphasis on correct methodology and consistency.

Volunteer

PyTorch Tester & Mentor Remote
DeepLearning.AI 2025.08 - 2026.02

- **Bug Hunting**: Test courses that includes lectures, quizzes, labs and assignments for general and code specific bugs, and report them.
- **Mentorship**: Guide learners on PyTorch and deep learning, resolving quiz conflicts, clarifying doubts, and supporting forum discussions in DeepLearning.AI courses.

Instructor & Event Organizer Kathmandu, Nepal
Advanced College of Engineering and Management 2022.03 - 2023.01

- **Workshop**: Led hands-on workshops mentoring 100+ peers and juniors in programming and ML fundamentals.
- **Curriculum Development**: Developed a structured curriculum covering Python (basic to advanced) and introductory Machine Learning.
- **Techfest**: Organized international zonal qualification from Nepal for IITB Techfest.

References

Prof. Dr. Bal Krishna Bal
Associate Dean, Kathmandu University
Professor, Department of Computer Science and Engineering, School of Engineering
bal@ku.edu.np

Dr. Gabriel Recchia
Director, Modulo Research Ltd
gabe@moduloresearch.com

Prof. Dr. Subarna Shakya
Director, Information Technology Innovation Center, Tribhuvan University
Professor, Department of Electronics and Computer Engineering, Institute of Engineering
drss@ioe.edu.np